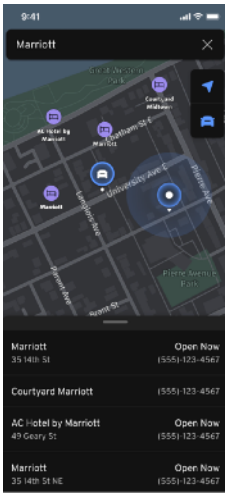
GITHUB.COM/FROZENPYROZEN | WWW.LINKEDIN.COM/IN/KYRYLO-KOHAN

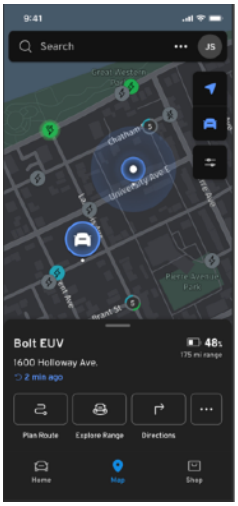
My Portfolio

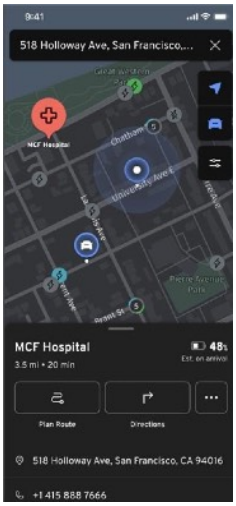
Kyrylo Kohan

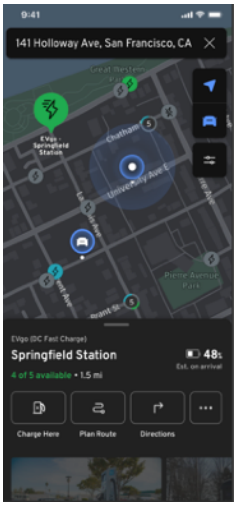
Car automotive management app under NDA General Motors(React Native)

Car automotive management app allow user to fully control his car from phone, there was features like running engine, get pressure in tears. User manage a lot of GM









Responsibilities

I was responsible for leading Map team of 8 developers (6 RN, 1 IOS, 1 Android) for huge and complex high load app for managing your car (ICE/EV), based on React Native and HereSDK and on reusable map provider module (display map/search/drawer/ markers/charging stations) RN micro frontend app, which integrated in multiple company productsCommunicated about dependencies with ui/backend(EV/ICE vehicle service, LBS service)/3rd part apps teams(HereSDK/HereApi) (giving scope/deadlines/priorities, leading plannings, solving blockers, creation of integration strategy, automation testing

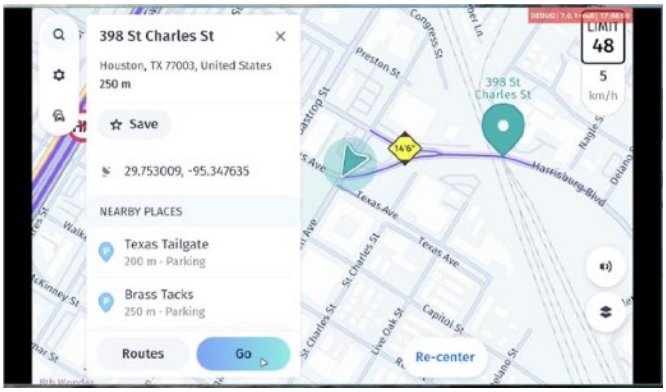
Created and documented process of delivering and testing our apps inside parent (parent app -> reusable feature app -> map provider app (RN +native IOS/Android + hereSdk))By feature request of client changed architecture of reusable feature app to migrated it into parent (parent app+ feature app -> map provider app (RN +native IOS/ Android + hereSdk)). Created plan how to migrate all logic and components from reusable feature to parent app according parent app patterns and practices (clean architecture, domain driven architecture, abstract services, pure components) During the migration changed reusable feature weak architecture points to speed up the integration and development processes(different map instances used global ref; map-provider map events listener was global and caused bugs when multiple instances were running. for each map separately; separated logic for map to map provider from reusable feature; designed an architecture for complex reused search functionality on the different screens, normalised state structure to store markers and search result)

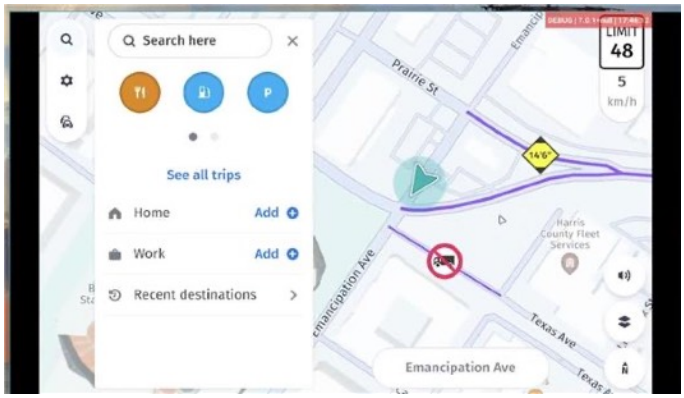
Technologies and frameworks

React native, HereSDK, Here api, Redux toolkit, Typescript, Azure, React hooks, git, Azure pipelines, jest, browser stack, domain driving design, clean architecture

Car automotive navigation app under NDA (React native/React, NodeJs, Android) Here (HERE Technologies)

Customer connected with package delivery which works based on Here solutions (hnod, Hnac, hereSDK, hereAPI). We got customsable Android app based on Js (NodeJs+React Native/React) and Android (Kotlin, Java) tools connected by gRPC. It's navigation app which installed on drivers tablet, based on hereMap but for large truck drivers. You could see route to desired place, restrictions (set your truck params like width, height) to show restrictions. These app could be extended by custom logic and features functionality based on service packages which connected to gRPC bundler and allows to add menu items, screens, popup, routing logic based on React Native/React + NodeJs. Generally project structure is hnod library + service packages + Android native/Web app.





Responsibilities

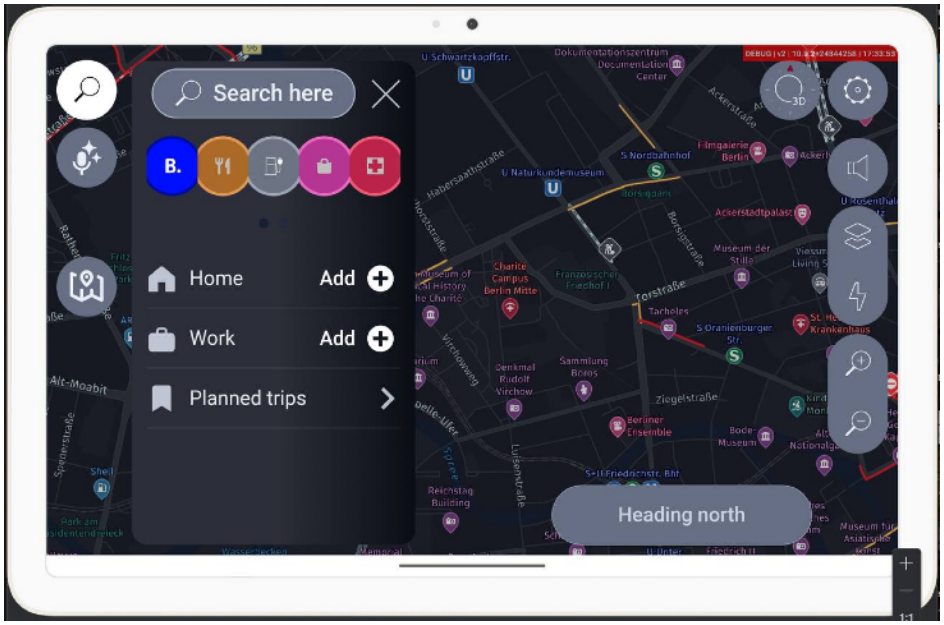
I was responsible for handling truck driver deviations where he goes off from desired route, collect such cases and sent them to report endpoint. I worked in Here Hnod library to extend functionality with adding route changes events submitting (New route started, Previous route restored, Route edited, Alternative route selected,Driver deviated from route, Route refresh, Driver parking). Then I'm handling needed events and comparing route data to understand if there was a deviation and sending that data to customer

Technologies and frameworks

Javascript, Typescript, NodeJs, gRPC, React Native, React, custom Here solutions (hnod library, here bundler, here runner, here api, here sdk, hnav library), RxJs, jest, Gitlab pipelines

Car navigation PoC (Node.JS, React) NDA

Automotive app designed to show possibilities of latest HERE technology on Volkswagen AG cars (Audi, Porsche, Volkswagen, etc.) and integrate latest features (Hugo AI assistant, L3 interaction, ADAS, Kurviger, ODD, custom theme, layout support etc.



Responsibilities

Implemented L3 interaction feature, Hugo AI assistant feature to build routes and display nearby places, Porsche theme and layout implementation, adding debug car data settings for EV cars, created script to optimise inefficient manual steps to make a build faster 3x

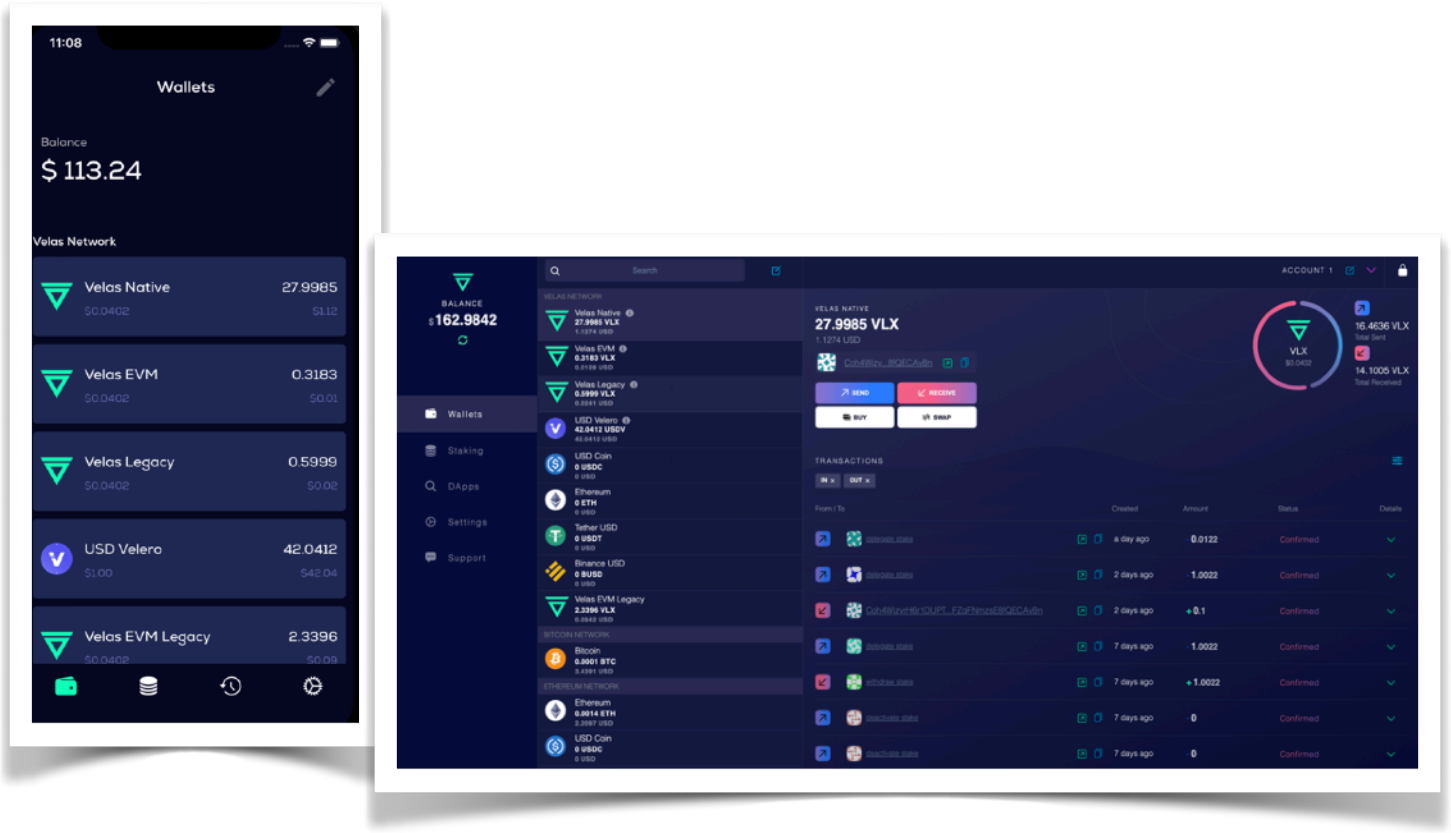
Technologies and frameworks

Javascript, Node.JS, Typescript, React, Rx.Js, gRPC, Kotlin, Java, Hnav, Hnod, Hugo AI, L3, Kurviger, protobuf.js.

Velas Wallet (React/React Native/Electron)

Fully open source **non-custodial** multi-language crypto wallet which supports Velas, Ethereum, BSC, Heco, BTC networks. You can stake VLX and earn rewards, swap tokens from one network to another via bridges, send/receive different crypto by QR code.

Includes web, mobile and desktop version.



Responsibilities

Elimination of problem areas with performance in wallet, reduced downloading speed of staking page loading from 2 mins to 5sec, get rid from extra re-renders of components for improving app usability and performance, optimised responses from BE side to improve speed removed unusable fields, added filters and paginations, configured building and signing on CI/CD on GitHub actions for wallets, added e2e testing for mobile wallet with detox, deployment app to F-droid and snap stores, designing architecture of new wallet, set up unit tests on jest

Technologies and frameworks

React-native-web, React native, Web3, Mobx, Typescript, Github Actions, React hooks, git, Expo eject, rn-cli, Javascript, Livescript, jest.

Jira design generation plugin (Node.JS, React, Android) NDA

This **Generative AI application** is designed to drastically accelerate the workflow for designers, enabling them to rapidly create **MVP, Proof of Concept (PoC), and presale designs** for customers. The tool offers seamless integration with the client's **Jira board**. Users can input their requirements via text prompts, provide a URL link, and upload specific photos or style schemas. The system then processes this input to instantly generate a design (e.g., in a Figma format), automatically **creates a new Jira ticket** detailing the request, and returns the finished design, thus streamlining the initial design and project planning phases.

Responsibilities

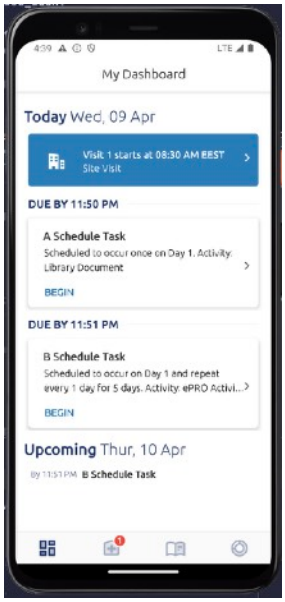
Engineered a specialized Generative AI Jira Plugin using **Jira Forge Custom UI** and **Atlaskit** with **TypeScript** to automate website design generation. A key architectural challenge was **bypassing Jira Forge's strict file upload limitations**; this was achieved by integrating **Azure Blob Storage** to manage chunked image data transfer. The solution provided full end-to-end functionality, including **interacting with the Jira API** to automatically create tickets with the generated design and specifications, with successful and **on-time project delivery** realized through effective collaboration with **Backend and ML teams**.

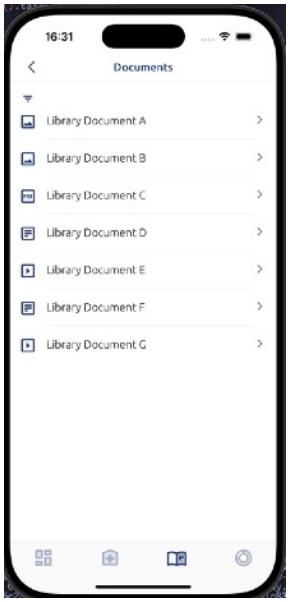
Technologies and frameworks

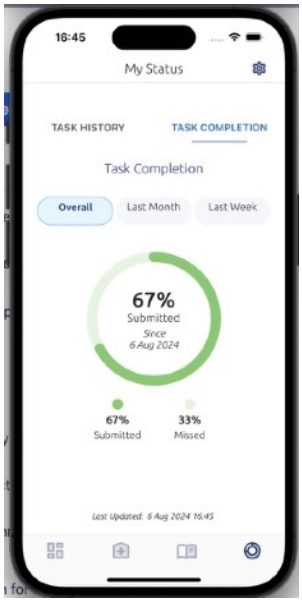
Javascript, Node.JS, Typescript, React, Jira, Atlaskit, Forge CustomUI

Clinical Inc app (React native, React, Android) Clinical inc

App designed to make it easier to be a participant in a clinical trial. Depending on the clinical trial, Clinical ink may provide some or all of the following capabilities: · Assistance with addresses and/or directions for your upcoming visits · Reminders to take medications or perform tasks that are part of your clinical trial · Relevant videos or content to help you understand more about the trial that you are part of, or to help you prepare for upcoming tasks or appointments · Questionnaires about how you are feeling etc. trial tasks ·







Responsibilities

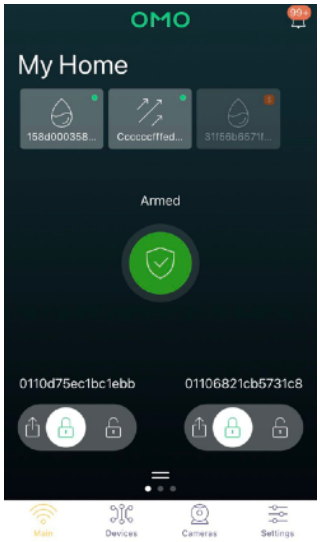
Refactored and optimized the core application flow, integrating it with the patient document library for company healthcare app. Ensured full unit test coverage using TypeScript and modern React Hooks best practices to boost maintainability and performance. Migrated Android app from version 13 to 14 to comply with Google Play Store requirements. Upgraded key dependencies and resolved complex migration issues (e.g., net.zetetic:sqlcipher-android, SQLite, Android 14 compatibility).

Technologies and frameworks

Javascript, Typescript,, React Native, Expo, React, Kotlin, Sqlcipher, Redux toolkit, aws sdk s3, was cognito, GraphQl, i18next, Apollo, aws-amplify, bitbucket pipelines, Jenkins

Smart home eco-system IOT devices management app (React Native) OMO Systems

App provides access control, security & home automation built for the multi-apartment buildings for worldwide. Key features of application are:



1. **Access to apartment -** control and open any entrances and doors from a single app, it may be entrance doors, barriers, elevators, gates and garbage cans.
2. **Video surveillance -** observe objects and territories from a single app, it includes online access to cameras and video archive from them, syncing with motion sensors with scenarios.
3. **Security -** protect your apartment from thieves, damage from breakdowns and accidents, which includes traffic monitoring, doors and windows opening, video surveillance in apartments, automatic shut-off of gas and water in case of leaks.

Also you can set up scenarios when you can trigger some devices on events, like start camera when motion sensor detect movement etc. You can manage your house, share it with family and enable face recognition for your cameras in your house. About security you can set all house for security guard and if there will be breakdown, there will be alert and auto call for security squad team.

Responsibilities

Feature owner of key monetization billing logic of application by tariff subscription, created SPA architecture and logic from scratch based on **React** for handling payments in web view of application and added blocking of unpaid devices and features on application. Implementing login by biometry with fixes of **react-native-keychain** library for Android Samsung device with FaceID and TouchID. fixed video streams issues in **react-native-vlcplayer** library with patches. Techlead for sprint of Frontend team.

Technologies and frameworks

React-native, Typescript , React navigation, React Native Navigation, React hooks, git, react-redux, reselect, redux-saga, Fastlane, react-native-vlcplayer, react-native-keychain, react-native-aws-cognito-js, react-native-webview, react-native-nfc-hce

IoT app with dosimeter (React Native) under NDA

This app helps doctors which works with radiation around the world to prevent overdose of it. There are three roles in the app: user, admin and superuser. Login process implemented by **Oauth2**. App consists of 10 screens and Rest API, Mqtt, **Bluetooth** requests. In general we read the information from radiation dosimeter by Bluetooth and send data to IoT server by **Mqtt** protocol. App can synchronize many dosimeters at the same time. There are main swipeable menu with bottom pagination with main screen where user can see progress of syncing, how many dosimeters synced with errors and time of sync. Then there are screen with showing details of current sync if sync is active now and the last sync if not. The last screen contains history of all syncs, which can be filtered by options from full screen modal. History stored in local **Realm** database. All secure values stored in **keychain.** Also app can update the dosimeter firmware. There are two menus in the left and right sides of the app, from where user can go to Profile, Settings and Logs(superuser only) page. On the filter modal user can sort syncs by date(from to), by username who wears it, by errors and id.

Responsibilities

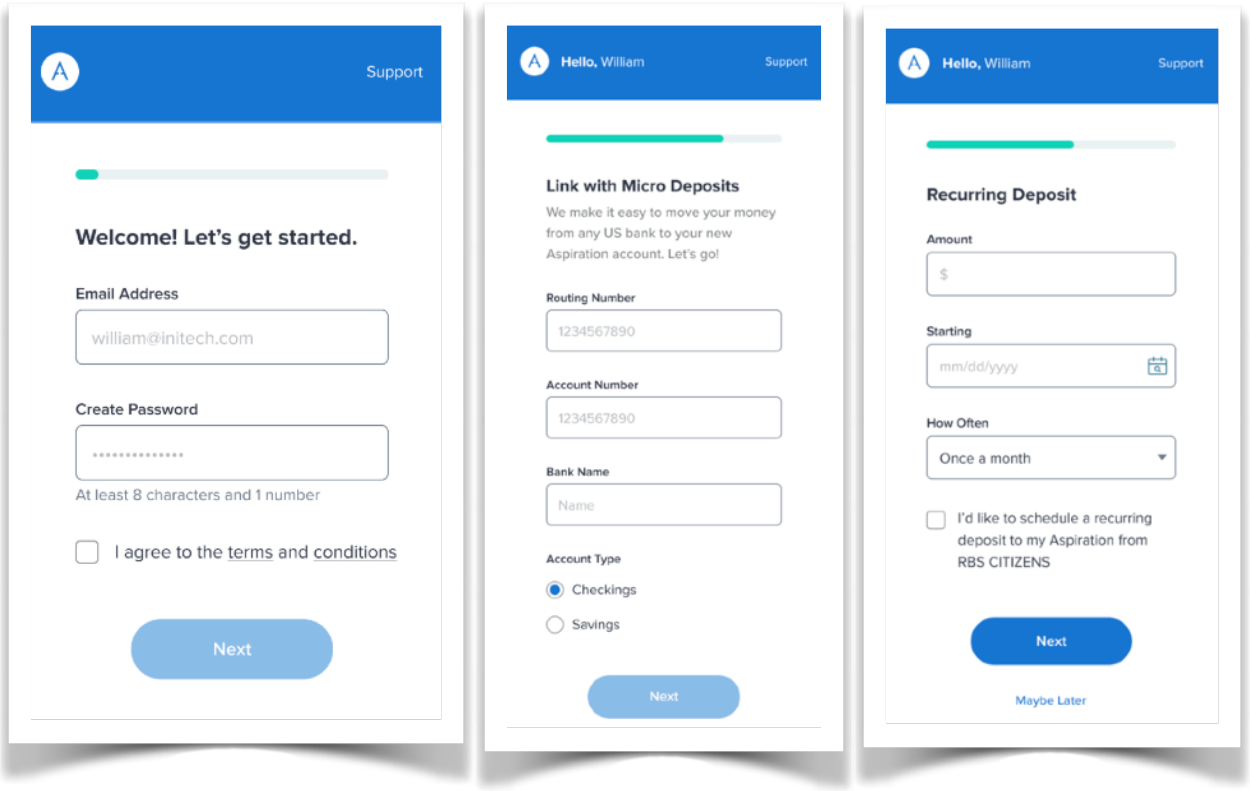
Initializing and creating API communication logic with UI screens for application, investigating, fixing and rewriting react-native and native libraries to work properly and serve our purposes. Active participation in meetings of the project, providing the most effective solutions for business, making a code reviews, adding logic for connecting to an app API, cover code with unit tests, working with **TDD** approach, **Mqtt logic** integration, creating login authorization flow with **Oauth2**, providing ideas of how to improve UI experience, working with background jobs on Android and IOS, improved dosimeter data parsing, which prevents overriding data, adding and creating Native Modules, setting up the **Realm** db connection and writing filter logic.

Technologies and frameworks

React-native, Typescript , React navigation, React hooks, git, Fastlane, react-native-ble-plx, Jira, react-native-app-auth, formik, @react-native-async-storage/async-storage, redux-saga, react-native-fs, react-native-keychain, react-native-native-mqtt, realm, eslint, husky

Aspiration signup(React Native Web)

Aspiration is a huge company which develop banking app ecosystem targeting on USA customers. It's a eco bank which help our planet by planting trees and investing money in reusable energy sources and fossil fuel. They moved signup flow from Angular to React.



Responsibilities

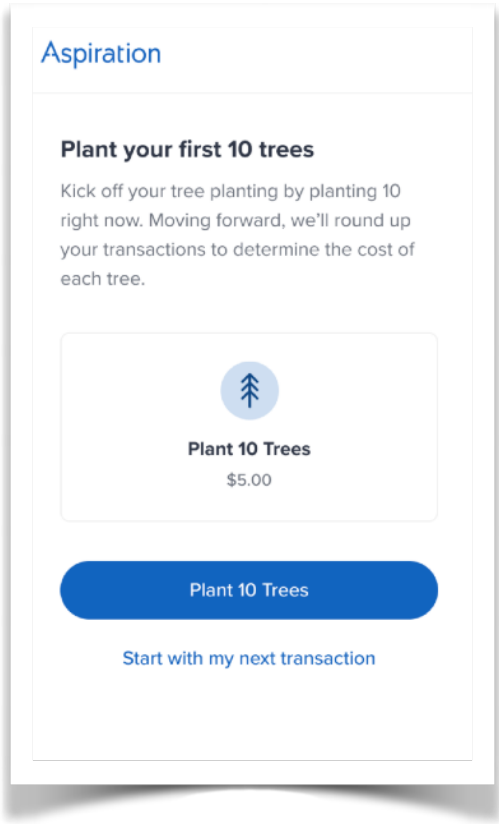
Fix and refactor an existing signup flow for application, add new screens and components. collaborating with native mobile developers for helping with react-native to native app integration, communication with American developers. Active participation in meetings of the project, providing the most effective solutions for business, making a code reviews, adding logic for connecting to an app API, cover code with unit tests, creation of e2e test utility, using CDD architecture across the project, Plaid/Google places/ReCaptcha v3/ integration, communication with product managers, manual react-native libraries linking, writing a PR style guide and documentation for easier onboarding, adding A/B tests and providing ideas of how to improve UI experience.

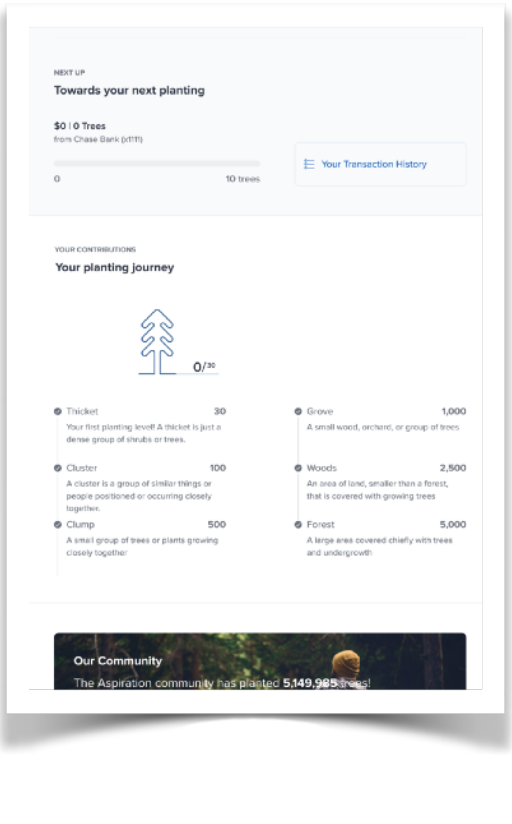
Technologies and frameworks

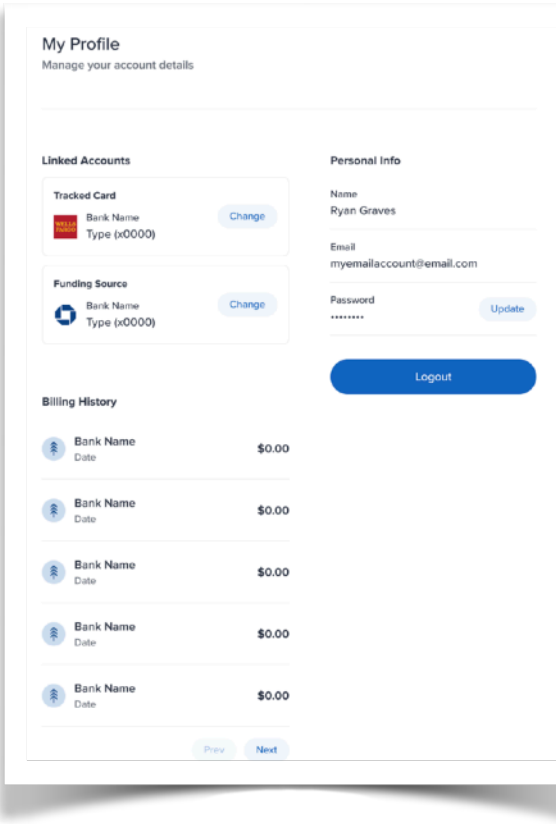
React-native-web, Apollo GraphQL, Typescript , React navigation, React hooks, git, Circle CI, Plaid, Cucumber, Puppeteer, Jest, Planaout, Segment, Rollbar, Crashlytics, Scaffolding, Smarty sheets, CDD, DDD, Next.js, Storybook, Eslint, CSS Grid, Jira

Aspiration PYC(React)

Flow for Banking web app of planting trees from your transaction, changing profile, settings etc. Currently in progress, so got not enough info about it.







Responsibilities

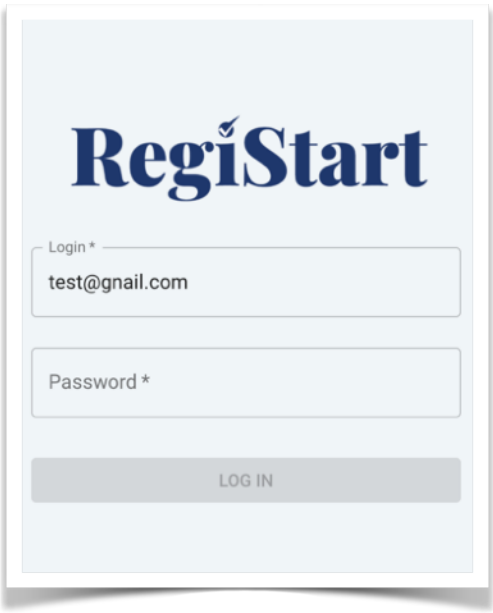
Create the application from scratch based on design, with mobile first approach, creating reusable components using Atomic design, added auto scaffolding file structure generator, communication with remote team from USA, active participation in grooming/ retrospectives of the project, solving communication problems between remote teams, planning scope of task for next period.

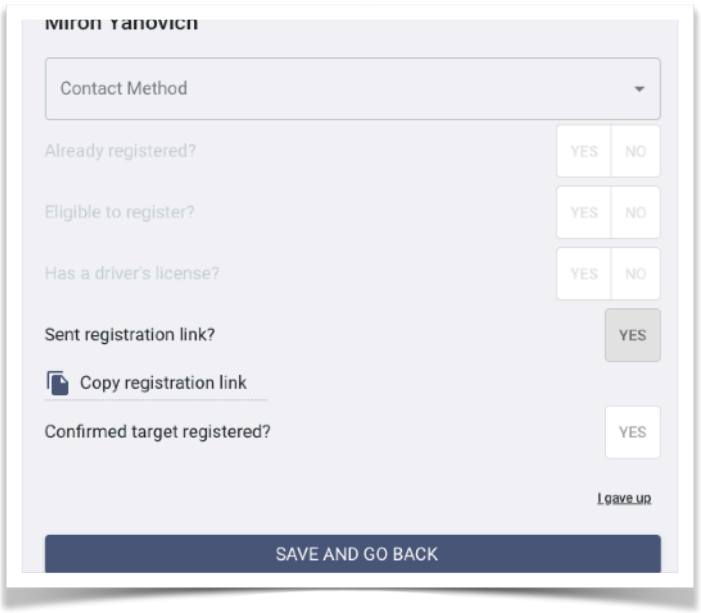
Technologies and frameworks

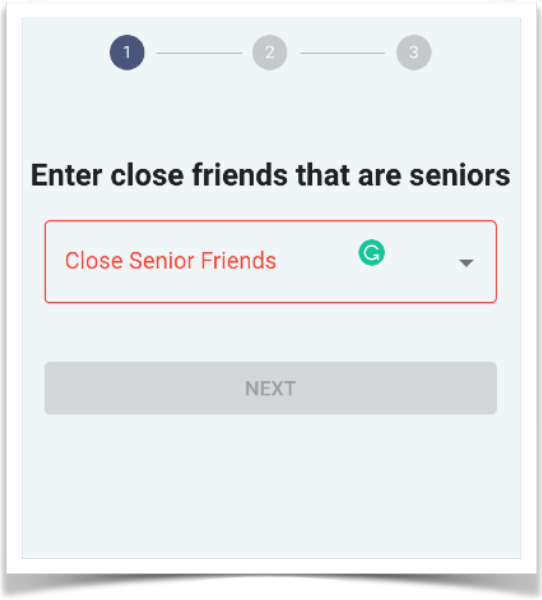
React, Apollo GraphQL, React hooks, git, Plaid, Jest, Atomic, Next.js, Storybook, Eslint, CSS Grid, Atomic design, CDD, UI Patterns, Jira, styled-components, Plop

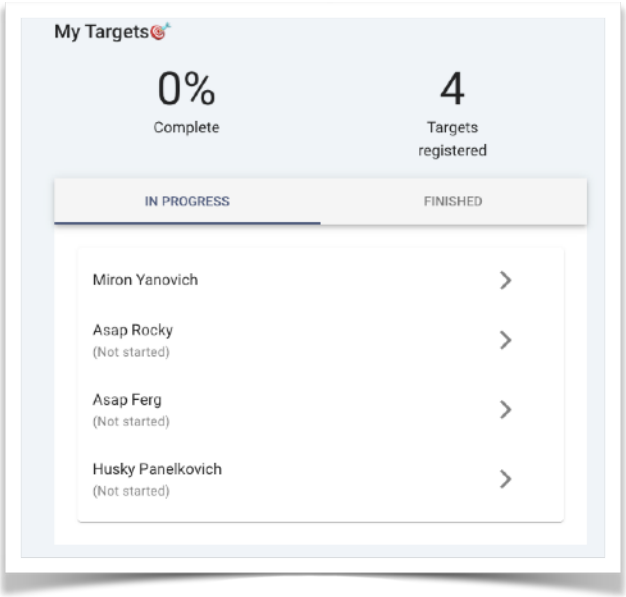
RegiStart(React Web, Node.js Backend, CI/CD)

RegiStart is a start-up non-profit that is leveraging the power of social networks and technology to systematically increase voter registration among 18-year-olds. The main goal of application is to increase in voter registration rates in schools using RegiStart by distributing seniors between student organisers which will motivate them to vote. Their slogan: “Strengthening America by registering voters, educating students on citizenship, and shaping students into citizens”.









Responsibilities

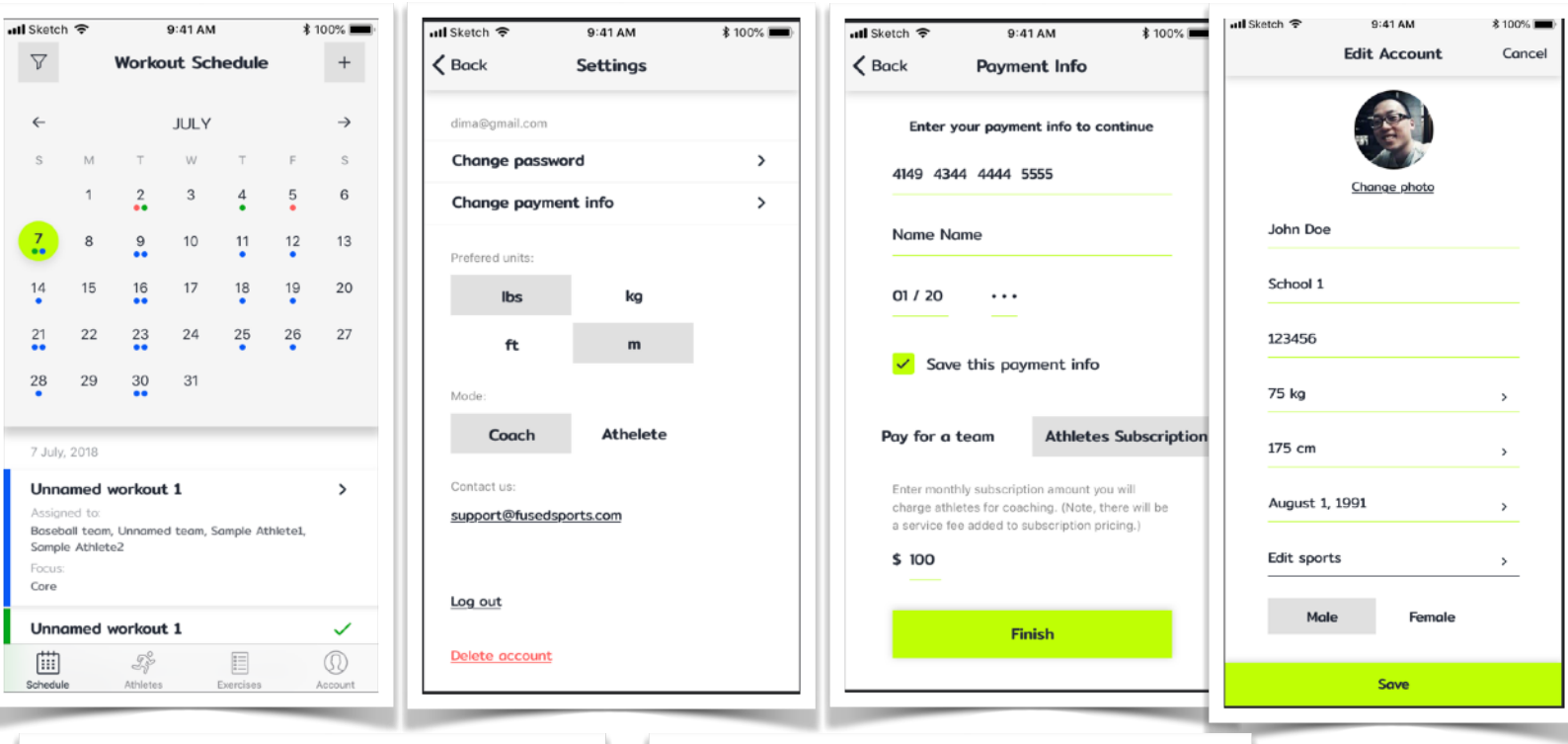
Development the MVP of frontend application from scratch based on sketches with React, MaterialUI, planning DB tables and relationships, creation of backend server for app on Node.Js with connection for DB using sequelize, adding DB on AWS RDS(postgresql), configuring CI/CD for deployment process on AWS EC2 and S3, adding migrations for DB with node-pg-migrate, creation of optimize distribution algorithm of students between organizers.

Technologies and frameworks

React, Redux, Typescript , AWS EC2 S3 RDS, CI/CD with Bitbucket pipelines, PostgreSQL, sequelize, node-pg-migrate, Formik, React Material UI, Ngnix, pm2, Jira

Fused(Web, mobile and backend)

The application has two roles - Coach and Athlete. Coach can invite athletes, create workouts, exercises, for athletes. User can change his role on settings screen Also create a teams from athletes and check their workout progress. For creating subscription coach should add card information and bank account hosted. App was published in Google play



Responsibilities

Add new functionality in application, work with API, creating flow for Athlete in Web version(previously it was only for Coach), moving all payment logic from mobile into Web version and added some endpoints on backend for removing card, making the web version fully responsible, fixing existing bugs on mobile. Web and backend side, added migration of dev and prod DB to AWS RDS, fully configured CI/CD process and created dev backend EC2 instance. Creating documentation . Direct communication with client

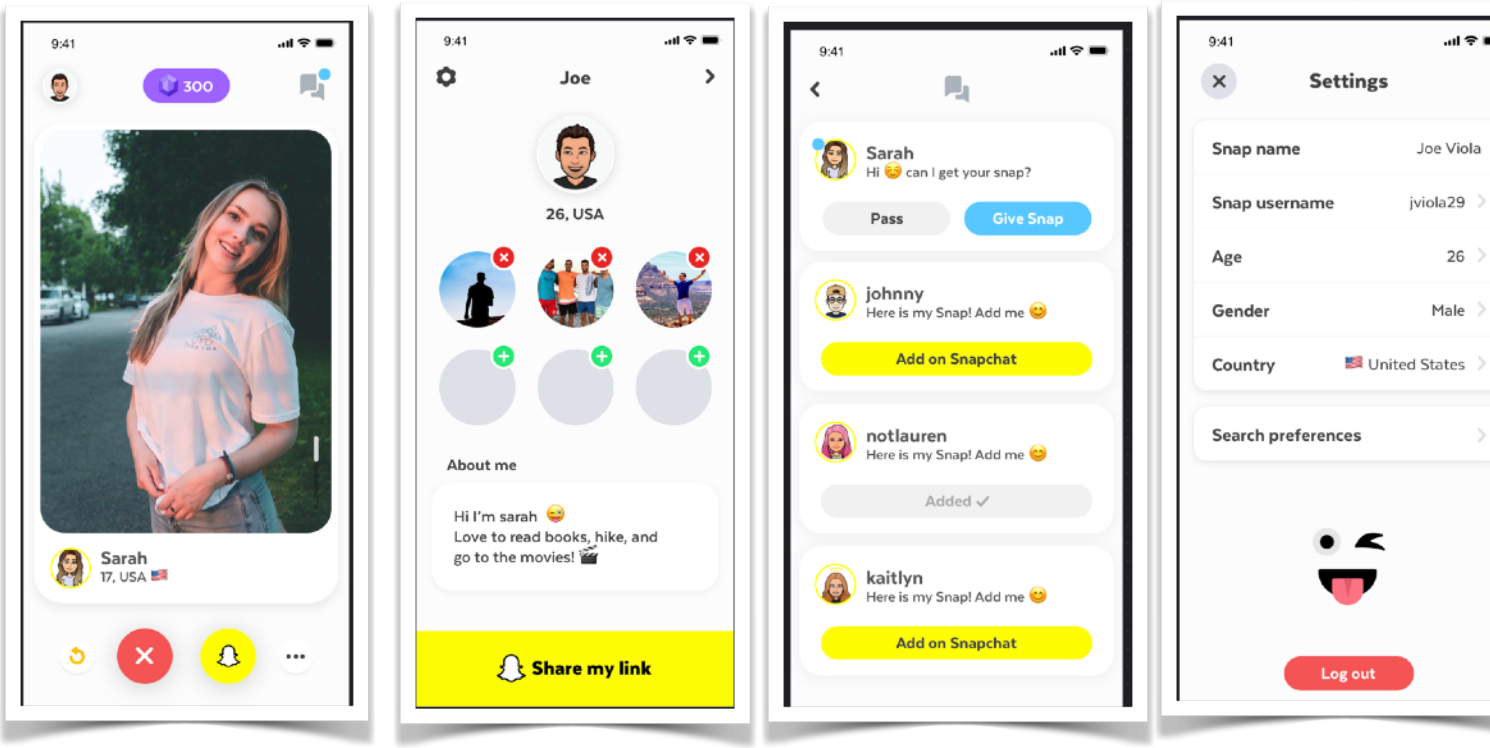
Technologies and frameworks

React native, React, React router, React navigation, NodeJs, sequelize, PostgreSQL, React hooks, Redux, RxJs, Stripe, Docker, AWS, Bugsnag, Bitbucket Pipelines, Fastlane, Git, Ngnix, pm2, Jira

Wink(React Native)

The application has a user, which can like/dislike others by swiping left/right for Gems(game currency), get notifications that he was liked by others, change his profile settings, and also earn gems by making actions like share link app with friends, claim daily price and left comment about app in google play. The app was successfully published in Google Play and App Store and connected more than 5 minion users.

Few screens of application:



Responsibilities

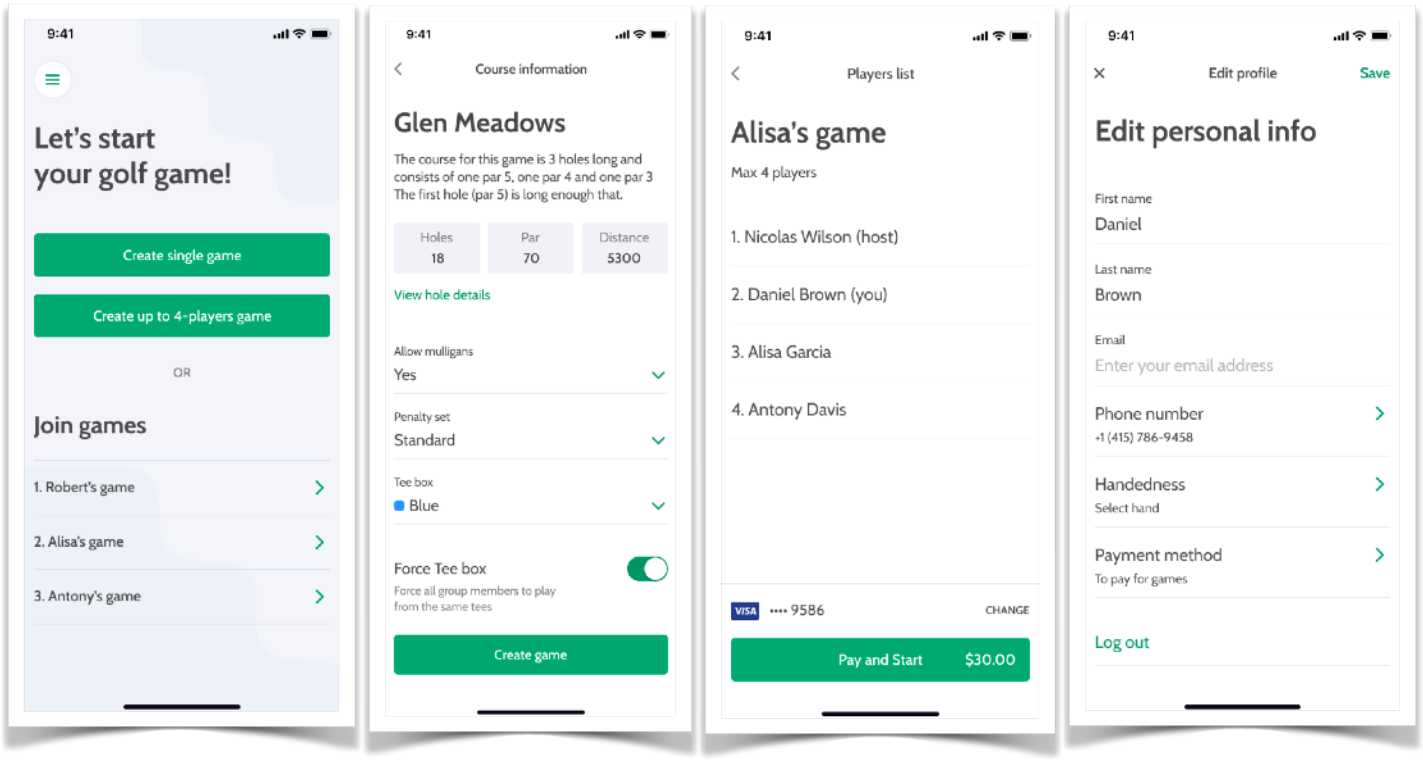
Development of application from scratch based on design, login using Snapchat/Mobile number, integration with BackendAPI, adding push notifications, Optimization and testing, configuring of CI/CD deployment, adding image optimize logic on backend side on Node.js, direct communication with customer. Making a code reviews

Technologies and frameworks

React native, Typescript , React hooks, React navigation , Redux, Thunk, Firebase, Bugsnag, Fastlane, Bitrise, AWS, NodeJs, git, CSS Grid, Flexbox, Jira

Golfuture(React native)

Golfuture is a proprietary new concept of golf that allows players to enjoy many of the best elements of traditional golf in a more time-efficient, eco-friendly and affordable way. Players use our smartphone application to select a course and their playing partners, and are guided through the facility at their own pace—no waiting on others. We have combined innovative course design with modern ball-tracking technology to create an exciting environment for golfers. App was published in Google play and app store.



Responsibilities

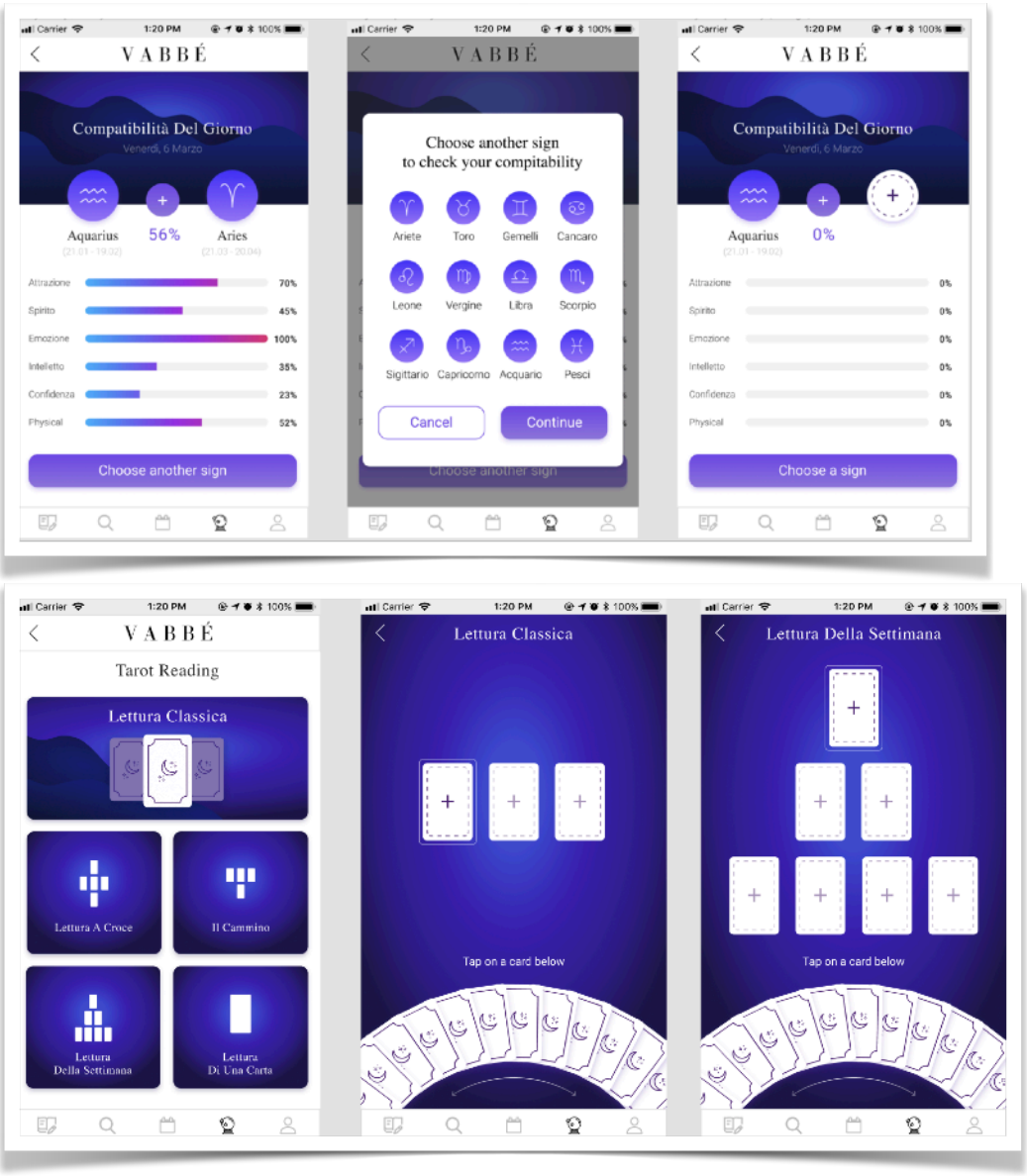
Development of application from scratch based on design, planning and creating Components and pages, adding business logic and add connection for to Stripe and different API's. Optimization and testing of the application, active participation in meetings and plannings of the project. Directly presentation the results of job for customer and communication with him.

Technologies and frameworks

React native, Redux, Typescript , Thunk, React navigation , Fastlane, Bugsnag, React hooks, Stripe, Bitrise, git, CSS Grid, Jira

Oroscopo(React Native)

Oroscopo is a tarot, astrology mobile app which can make predictions for zodiac signs, make different tarot layouts for future and calculating a zodiac compatibility between signs. App was published in Google play and app store



Responsibilities

Add new functionality in application, work with app API, adding UI components, animated card circle, zodiac compatibility logic, add banners integration and creating new screens with tarot layouts. Optimization and testing of the application, active participation in meetings of the project, directly presentation the results of job for customer.

Technologies and frameworks

React native, Redux, Typescript , Thunk, React navigation , React hooks, git, Firebase, AdMob, Jira